

MERN stack - Shopping Online

Lab 02

Contents

1	MERN stack	2
1.0.1	Models	2
1.0.2	Utils	2
1.0.3	Stylesheets	4
1.1	Functionals	5
1.1.1	Admin - login & logout	5

1 MERN stack

1.0.1 Models

- Create 'server/models/Models.js' file

```
1 //CLI: npm install mongoose --save
2 const mongoose = require('mongoose');
3 // schemas
4 const AdminSchema = mongoose.Schema({
5   _id: mongoose.Schema.Types.ObjectId,
6   username: String,
7   password: String
8 }, { versionKey: false });
9 const CategorySchema = mongoose.Schema({
10   _id: mongoose.Schema.Types.ObjectId,
11   name: String
12 }, { versionKey: false });
13 const CustomerSchema = mongoose.Schema({
14   _id: mongoose.Schema.Types.ObjectId,
15   username: String,
16   password: String,
17   name: String,
18   phone: String,
19   email: String,
20   active: Number,
21   token: String,
22 }, { versionKey: false });
23 const ProductSchema = mongoose.Schema({
24   _id: mongoose.Schema.Types.ObjectId,
25   name: String,
26   price: Number,
27   image: String,
28   cdate: Number,
29   category: CategorySchema
30 }, { versionKey: false });
31 const ItemSchema = mongoose.Schema({
32   product: ProductSchema,
33   quantity: Number
34 }, { versionKey: false, _id: false });
35 const OrderSchema = mongoose.Schema({
36   _id: mongoose.Schema.Types.ObjectId,
37   cdate: Number,
38   total: Number,
39   status: String,
40   customer: CustomerSchema,
41   items: [ItemSchema]
42 }, { versionKey: false });
43 // models
44 const Admin = mongoose.model('Admin', AdminSchema);
45 const Category = mongoose.model('Category', CategorySchema);
46 const Customer = mongoose.model('Customer', CustomerSchema);
47 const Product = mongoose.model('Product', ProductSchema);
48 const Order = mongoose.model('Order', OrderSchema);
49 module.exports = { Admin, Category, Customer, Product, Order };
```

1.0.2 Utils

- Create 'server/Utils/MyConstants.js' file

```
1 const MyConstants = {
2   DB_SERVER: '<cluster>.mongodb.net',
3   DB_USER: '<db_user>',
4   DB_PASS: '<db_pass>',
5   DB_DATABASE: '<db_name>',
6   EMAIL_USER: '<email_user>', // Microsoft mail service
7   EMAIL_PASS: '<email_pass>',
8   JWT_SECRET: '<jwt_secret>',
```

```

9   JWT_EXPIRES: '<jwt_expires>', // in milliseconds
10 };
11 module.exports = MyConstants;

```

- Create 'server/utils/MongooseUtil.js' file

```

1 //CLI: npm install mongoose --save
2 const mongoose = require('mongoose');
3 const MyConstants = require('./MyConstants');
4 const uri = 'mongodb+srv://' + MyConstants.DB_USER + ':' + MyConstants.DB_PASS + '@' +
   MyConstants.DB_SERVER + '/' + MyConstants.DB_DATABASE;
5 mongoose.connect(uri, { useNewUrlParser: true })
6   .then(() => { console.log('Connected to ' + MyConstants.DB_SERVER + '/' + MyConstants.
   DB_DATABASE); })
7   .catch((err) => { console.error(err); });

```

- Create 'server/utils/CryptoUtil.js' file

```

1 //CLI: npm install crypto --save
2 const CryptoUtil = {
3   md5(input) {
4     const crypto = require('crypto');
5     const hash = crypto.createHash('md5').update(input).digest('hex');
6     return hash;
7   }
8 };
9 module.exports = CryptoUtil;

```

- Create 'server/utils/EmailUtil.js' file

```

1 //CLI: npm install nodemailer --save
2 const nodemailer = require('nodemailer');
3 const MyConstants = require('./MyConstants');
4 const transporter = nodemailer.createTransport({
5   service: 'hotmail',
6   auth: {
7     user: MyConstants.EMAIL_USER,
8     pass: MyConstants.EMAIL_PASS
9   }
10 });
11 const EmailUtil = {
12   send(email, id, token) {
13     const text = 'Thanks for signing up, please input these informations to activate your
   account:\n\t .id: ' + id + '\n\t .token: ' + token;
14     return new Promise(function (resolve, reject) {
15       const mailOptions = {
16         from: MyConstants.EMAIL_USER,
17         to: email,
18         subject: 'Signup | Verification',
19         text: text
20       };
21       transporter.sendMail(mailOptions, function (err, result) {
22         if (err) reject(err);
23         resolve(true);
24       });
25     });
26   }
27 };
28 module.exports = EmailUtil;

```

- Create 'server/utils/JwtUtil.js' file

```

1 //CLI: npm install jsonwebtoken --save
2 const jwt = require('jsonwebtoken');
3 const MyConstants = require('./MyConstants');
4 const JwtUtil = {
5   genToken(username, password) {
6     const token = jwt.sign(
7       { username: username, password: password },
8       MyConstants.JWT_SECRET,

```

```

9      { expiresIn: MyConstants.JWT_EXPIRES }
10    });
11    return token;
12  },
13  checkToken(req, res, next) {
14    const token = req.headers['x-access-token'] || req.headers['authorization'];
15    if (token) {
16      jwt.verify(token, MyConstants.JWT_SECRET, (err, decoded) => {
17        if (err) {
18          return res.json({
19            success: false,
20            message: 'Token is not valid'
21          });
22        } else {
23          req.decoded = decoded;
24          next();
25        }
26      });
27    } else {
28      return res.json({
29        success: false,
30        message: 'Auth token is not supplied'
31      });
32    }
33  }
34 };
35 module.exports = JwtUtil;

```

1.0.3 Stylesheets

- Update 'client-admin/src/App.css' and 'client-customer/src/App.css' files

```

1  /* styles.css */
2  .body-customer {
3    width: 1150px;
4    height: fit-content;
5    margin: 0px auto;
6  }
7  .body-admin {
8    width: fit-content;
9    height: fit-content;
10   margin: 0px auto;
11 }
12 div.inline {
13   display: inline-block;
14 }
15 figure.caption-right {
16   display: flex;
17   align-items: center;
18 }
19 .align-center {
20   width: fit-content;
21   margin: 0px auto;
22 }
23 .align-valign-center {
24   position: absolute;
25   top: 50%;
26   left: 50%;
27   transform: translate(-50%, -50%);
28   padding: 0px 20px 20px 20px;
29   border: 1px solid black;
30 }
31 .text-center {
32   text-align: center;
33 }
34 span.link {
35   cursor: pointer;
36   color: blue;

```

```

37     text-decoration: underline;
38 }
39 /* menu.css */
40 div.border-bottom {
41     border-bottom: 1px solid black;
42     padding: 10px 0px 10px 0px;
43 }
44 div.float-left {
45     float: left;
46 }
47 div.float-right {
48     float: right;
49 }
50 div.float-clear {
51     clear: both;
52 }
53 ul.menu {
54     margin: 0px;
55     padding: 0px;
56 }
57 li.menu {
58     display: inline;
59     margin-right: 20px;
60     text-transform: uppercase;
61 }
62 form.search {
63     margin: 0px;
64 }
65 input.keyword {
66     width: 110px;
67 }
68 /* datatable.css */
69 .datatable {
70     text-align: center;
71     border-collapse: collapse;
72 }
73 /* header */
74 tr:first-child.datatable {
75     background-color: #ffcc00;
76 }
77 /* even rows, except header */
78 tr:not(:first-child):nth-child(even).datatable {
79     background-color: #f5f5f5;
80 }
81 /* odd rows, except header */
82 tr:not(:first-child):nth-child(odd).datatable {
83     background-color: #ffeeaa;
84 }
85 /* except header */
86 tr:not(:first-child):hover.datatable {
87     background-color: #dcfcfc;
88     cursor: pointer;
89 }

```

1.1 Functionals

1.1.1 Admin - login & logout

1.1.1.1 Server

- Create 'server/models/AdminDAO.js' file

```

1 require('../utils/MongooseUtil');
2 const Models = require('../Models');
3
4 const AdminDAO = {
5     async selectByUsernameAndPassword(username, password) {

```

```

6     const query = { username: username, password: password };
7     const admin = await Models.Admin.findOne(query);
8     return admin;
9   }
10 };
11 module.exports = AdminDAO;

```

- Create 'server/api/admin.js' file

```

1  const express = require('express');
2  const router = express.Router();
3  // utils
4  const JwtUtil = require('../utils/JwtUtil');
5  // daos
6  const AdminDAO = require('../models/AdminDAO');
7  // login
8  router.post('/login', async function (req, res) {
9    const username = req.body.username;
10   const password = req.body.password;
11   if (username && password) {
12     const admin = await AdminDAO.selectByUsernameAndPassword(username, password);
13     if (admin) {
14       const token = JwtUtil.genToken();
15       res.json({ success: true, message: 'Authentication successful', token: token });
16     } else {
17       res.json({ success: false, message: 'Incorrect username or password' });
18     }
19   } else {
20     res.json({ success: false, message: 'Please input username and password' });
21   }
22 });
23 router.get('/token', JwtUtil.checkToken, function (req, res) {
24   const token = req.headers['x-access-token'] || req.headers['authorization'];
25   res.json({ success: true, message: 'Token is valid', token: token });
26 });
27 module.exports = router;

```

- Update 'server/index.js' file

```

1  ...
2  // apis
3  app.use('/api/admin', require('./api/admin.js'));

```

- Postman:

- (POST) http://localhost:3000/api/admin/login
 - Body (raw+JSON): { "username": "admin", "password": "123" }
- (GET) http://localhost:3000/api/admin/token
 - Headers: "x-access-token": <token>

1.1.1.2 Client-admin

- Create 'client-admin/src/contexts/MyContext.js' file

```

1  import React from 'react';
2  const MyContext = React.createContext();
3  export default MyContext;

```

- Create 'client-admin/src/contexts/MyProvider.js' file

```

1  import React, { Component } from 'react';
2  import MyContext from './MyContext';
3
4  class MyProvider extends Component {
5    constructor(props) {
6      super(props);

```

```

7     this.state = { // global state
8         // variables
9         token: '',
10        username: '',
11        // functions
12        setToken: this.setToken,
13        setUsername: this.setUsername
14    };
15 }
16 setToken = (value) => {
17     this.setState({ token: value });
18 }
19 setUsername = (value) => {
20     this.setState({ username: value });
21 }
22 render() {
23     return (
24         <MyContext.Provider value={this.state}>
25             {this.props.children}
26         </MyContext.Provider>
27     );
28 }
29 }
30 export default MyProvider;

```

- Update 'client-admin/src/App.js' file

```

1 import './App.css';
2 import React, { Component } from 'react';
3 import MyProvider from './contexts/MyProvider';
4 import Login from './components/LoginComponent';
5 import Main from './components/MainComponent';
6
7 class App extends Component {
8     render() {
9         return (
10             <MyProvider>
11                 <Login />
12                 <Main />
13             </MyProvider>
14         );
15     }
16 }
17 export default App;

```

- Create 'client-admin/src/components/LoginComponent.js' file

```

1 import axios from 'axios';
2 import React, { Component } from 'react';
3 import MyContext from '../contexts/MyContext';
4
5 class Login extends Component {
6     static contextType = MyContext; // using this.context to access global state
7     constructor(props) {
8         super(props);
9         this.state = {
10             txtUsername: '',
11             txtPassword: ''
12         };
13     }
14     render() {
15         if (this.context.token === '') {
16             return (
17                 <div className="align-valign-center">
18                     <h2 className="text-center">ADMIN LOGIN</h2>
19                     <form>
20                         <table className="align-center">
21                             <tbody>
22                                 <tr>
23                                     <td>Username</td>

```

```

24         <td><input type="text" value={this.state.txtUsername} onChange={(e) => { this
25             .setState({ txtUsername: e.target.value }) }} /></td>
26     </tr>
27     <tr>
28         <td>Password</td>
29         <td><input type="password" value={this.state.txtPassword} onChange={(e) => {
30             this.setState({ txtPassword: e.target.value }) }} /></td>
31     </tr>
32     <tr>
33         <td></td>
34         <td><input type="submit" value="LOGIN" onClick={(e) => this.btnLoginClick(e)}
35             /></td>
36     </tr>
37 </tbody>
38 </table>
39 </form>
40 </div>
41 );
42 }
43 return (<div />);
44 }
45 // event-handlers
46 btnLoginClick(e) {
47     e.preventDefault();
48     const username = this.state.txtUsername;
49     const password = this.state.txtPassword;
50     if (username && password) {
51         const account = { username: username, password: password };
52         this.apiLogin(account);
53     } else {
54         alert('Please input username and password');
55     }
56 }
57 // apis
58 apiLogin(account) {
59     axios.post('/api/admin/login', account).then((res) => {
60         const result = res.data;
61         if (result.success === true) {
62             this.context.setToken(result.token);
63             this.context.setUsername(account.username);
64         } else {
65             alert(result.message);
66         }
67     });
68 }
69 }
70 export default Login;

```

- Create 'client-admin/src/components/MenuComponent.js' file

```

1  import React, { Component } from 'react';
2  import MyContext from '../contexts/MyContext';
3
4  class Menu extends Component {
5      static contextType = MyContext; // using this.context to access global state
6      render() {
7          return (
8              <div className="border-bottom">
9                  <div className="float-left">
10                     <ul className="menu">
11                         <li className="menu"><a href="#">Home</a></li>
12                         <li className="menu"><a href="#">Category</a></li>
13                         <li className="menu"><a href="#">Product</a></li>
14                         <li className="menu"><a href="#">Order</a></li>
15                         <li className="menu"><a href="#">Customer</a></li>
16                     </ul>
17                 </div>
18                 <div className="float-right">
19                     Hello <b>{this.context.username}</b> | <a href="#" onClick={() => this.lnkLogoutClick

```

```

20         {}>Logout</a>
21     </div>
22     <div className="float-clear" />
23 </div>
24 );
25 }
26 // event-handlers
27 lnkLogoutClick() {
28     this.context.setToken('');
29     this.context.setUsername('');
30 }
31 }
32 export default Menu;

```

- Create 'client-admin/src/components/HomeComponent.js' file

```

1 import React, { Component } from 'react';
2
3 class Home extends Component {
4     render() {
5         return (
6             <div className="align-center">
7                 <h2 className="text-center">ADMIN HOME</h2>
8                 
9             </div>
10         );
11     }
12 }
13 export default Home;

```

- Create 'client-admin/src/components/MainComponent.js' file

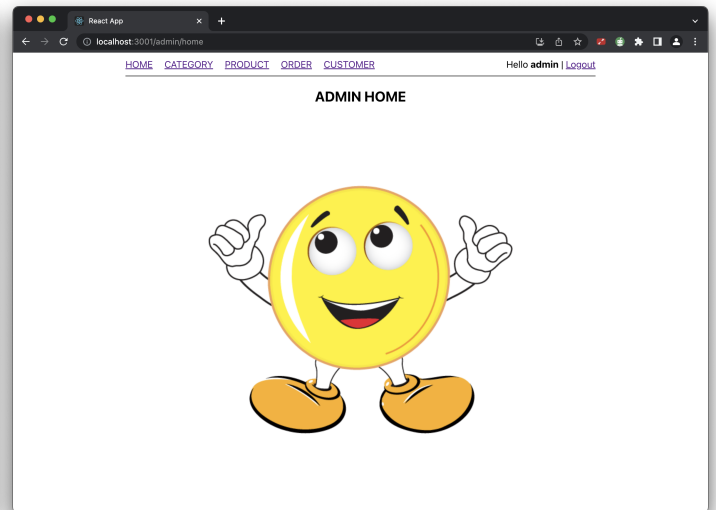
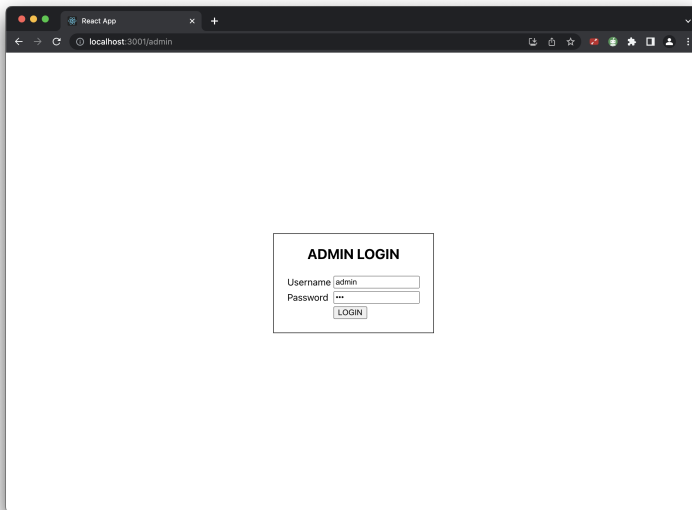
```

1 import React, { Component } from 'react';
2 import MyContext from '../contexts/MyContext';
3 import Menu from './MenuComponent';
4 import Home from './HomeComponent';
5
6 class Main extends Component {
7     static contextType = MyContext; // using this.context to access global state
8     render() {
9         if (this.context.token !== '') {
10             return (
11                 <div className="body-admin">
12                     <Menu />
13                     <Home />
14                 </div>
15             );
16         }
17         return (<div />);
18     }
19 }
20 export default Main;

```

- Browser: <http://localhost:3001/admin>

1.1.1.3 Screenshots



1.1.1.4 React Router

- Install react-router-dom library.

```
1 $ npm install react-router-dom --save
```

- Update 'client-admin/src/App.js' file

```
1 ...
2 import { BrowserRouter } from 'react-router-dom';
3
4 class App extends Component {
5   render() {
6     return (
7       ...
8       <BrowserRouter >
9         <Main />
10      </BrowserRouter>
11      ...
12    );
13  }
14 }
15 ...
```

- Update 'client-admin/src/components/MainComponent.js' file

```
1 ...
2 import { Routes, Route, Navigate } from 'react-router-dom';
3
4 class Main extends Component {
5   ...
6   render() {
7     ...
8     return (
9       <div className="body-admin">
10        <Menu />
11        <Routes>
12          <Route path="/admin" element={<Navigate replace to="/admin/home" />} />
13          <Route path="/admin/home" element={<Home />} />
14        </Routes>
15      </div>
16    );
17    ...
18  }
19 }
20 ...
```

- Update 'client-admin/src/components/MenuComponent.js' file

```

1  ...
2  import { Link } from 'react-router-dom';
3
4  class Menu extends Component {
5    ...
6    render() {
7      return (
8        ...
9        <div className="float-left">
10         <ul className="menu">
11           <li className="menu"><Link to="/admin/home">Home</Link></li>
12           <li className="menu"><Link to="/">Category</Link></li>
13           <li className="menu"><Link to="/">Product</Link></li>
14           <li className="menu"><Link to="/">Order</Link></li>
15           <li className="menu"><Link to="/">Customer</Link></li>
16         </ul>
17       </div>
18       <div className="float-right">
19         ... <Link to="/admin/home" onClick={() => this.lnkLogoutClick()}>Logout</Link>
20       </div>
21       ...
22     );
23   }
24   ...
25 }
26 ...

```

- Browser: <http://localhost:3001/admin>